

XGBoost Crop Yield Validity Analysis

Syed Ali
Texas State University
San Marcos, TX
s_a577@txstate.edu

Wajeeh Moonwar
Texas State University
San Marcos, TX
rfb47@txstate.edu

Gustavo Hernandez
Texas State University
San Marcos, TX
pow10@txstate.edu

Abstract—Agriculture relies on precise predictive modeling to mitigate climate risks and optimize agricultural resource allocation. This paper outlines the architecture of a machine learning framework designed to synthesize historical meteorological, edaphic, and agronomic data spanning the past two decades. The model evaluates the impact of soil composition and dynamic environmental stress factors on harvest outcomes. The predictive framework achieved an average R^2 value of 0.60 to 0.70 for Corn, with a Root Mean Square Error (RMSE) of 22 to 25 bushels per acre.

I. INTRODUCTION

Agriculture remains a fundamental pillar of global societal stability, necessitated by the continuous requirement for food and fiber production. Historically, agricultural decision-making relied upon heuristic techniques derived from generational trial and error. In the modern era, a significant portion of agricultural practitioners continues to depend on experiential intuition to manage crop cycles. However, the increasing volatility of global climate patterns renders subjective methodologies insufficient, necessitating the integration of robust machine learning (ML) architectures to mitigate risk and optimize yields. Current precision agriculture frequently leverages sophisticated platforms, such as Climate FieldView, which integrate historical yield metrics, real-time meteorological data, and high-resolution edaphic mapping to optimize field management. Mechanistic frameworks such as the Agricultural Production Systems Simulator (APSIM) and the Decision Support System for Agrotechnology Transfer (DSSAT) provide daily physiological simulations of plant growth; however, these high-compute models demand substantial computational resources and possess significant operational complexity.

While contemporary methodologies often utilize complex deep learning networks, tree-based machine learning algorithms such as Random Forest and XGBoost offer critical advantages. Early implementations of tree-based models were occasionally critiqued for treating data points in isolation, failing to account for cumulative biological stress. However, modern feature engineering allows XGBoost to offer an unparalleled speed-to-insight ratio and, crucially, model interpretability—enabling the precise identification of biological tipping points, such as specific thermal thresholds that induce crop failure.

This study implements an XGBoost framework focusing on the Texas region, synthesizing meteorological data from the NASA Prediction of Worldwide Energy Resources (POWER)

database with ground-truth yield records from the United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS). The analysis specifically examines Corn and Cotton due to their divergent physiological survival strategies and economic dominance. Corn exhibits extreme sensitivity to thermal stress during its determinate July tasseling phase, whereas Cotton, despite possessing greater heat hardiness, remains highly vulnerable to late-season precipitation. Critically, this research addresses the "Texas Paradox"—the statistical variance introduced by the coexistence of irrigated and dryland practices within the same datasets. This study establishes a computationally efficient framework to bridge the gap between biological field realities and reported agricultural statistics, explicitly accounting for phenomena such as crop abandonment resulting from extreme climatic events, such as the 2023 Heat Dome [3].

II. MATERIALS AND METHODS

A. Agronomical Data and Yield Compilation

The foundation of the predictive model relies upon the extraction of high-fidelity agricultural statistics. County-level yield data for Corn and Cotton in Texas were obtained from the USDA NASS database, encompassing the years 2000 through 2024. To resolve the statistical variance inherently present in regional agricultural data (the "Texas Paradox"), the datasets were systematically categorized by production practice, isolating irrigated yields from non-irrigated (dryland) profiles. This stratification is critical, as the climatic threshold for crop failure differs drastically between precipitation-reliant fields and those supplemented by aquifer extraction. Furthermore, historical planting progress metrics—expressed as the percentage of the crop planted per county by specific months—were integrated to provide an empirical timeline of crop establishment reflecting regional agronomic consensus.

B. Geospatial Mapping and Meteorological Data Extraction

To accurately align dynamic meteorological systems with static yield reports, precise geographic anchoring was necessitated. Geometric centroids (Latitude and Longitude) for each Texas county were utilized to query the NASA POWER database. Meteorological extraction was conducted on a monthly aggregated basis to capture macro-climatic trends. The comprehensive suite of retrieved environmental parameters included Average Temperature, Maximum and Minimum

Temperatures, Dew Point, Total Precipitation, Relative Humidity, Specific Humidity, Wind Speed, and All-Sky Surface Shortwave Downward Irradiance.

C. Temporal Windowing and Biological Alignment

The extraction of meteorological data was purposefully restricted to specific physiological growth windows to ensure the algorithm trained exclusively on variables impacting yield physiology. For Corn (determinate growth), the extraction window encompassed March through August. May and June represent critical vegetative and root establishment phases, while July encompasses the high-sensitivity tasseling period where extreme thermal events cause catastrophic pollen sterilization. For Cotton (indeterminate growth), the extraction window spanned May through October. July and August represent peak boll development, heavily dictated by thermal stress and moisture availability, while September and October parameters monitor late-season harvest conditions where excessive precipitation severely degrades fiber quality.

D. Edaphic Constraints

Static spatial baselines for the predictive models were established using edaphic data integrated from the USDA soil database. Key extracted parameters included topsoil clay percentage, soil pH, and soil organic carbon. The pH level is a critical determinant of crop viability; highly acidic environments (pH below 5.5) trigger chemical reactions that bind essential nutrients to clay particles, rendering them biologically unavailable to root systems [2]. Furthermore, excessive clay composition restricts root aeration, leading to waterlogging and oxygen deprivation [2]. Embedding these geological factors provided the model with a functional anchor to differentiate the baseline drought resilience between the heavy clays of the Blackland Prairies and the highly permeable loams of the High Plains.

E. Feature Engineering and Bio-Climatic Proxies

Raw meteorological variables frequently prove insufficient for modeling complex biological stress. To capture non-linear relationships between the environment and crop physiology, raw data were mathematically transformed into specific bio-climatic proxies. Vapor Pressure Deficit (VPD) was calculated to derive true atmospheric "drying power," which forces stomatal closure and halts photosynthesis during extreme heat. A normalized County-Level Yield Anomaly was engineered by measuring annual deviation from a county's historical average, allowing the model to evaluate weather-driven impacts independently of a region's static agricultural baseline. Additional cumulative metrics, including Harvest Rain Penalty and Heat Load, were incorporated to capture threshold-based thermal stress during reproductive windows.

III. RESULTS

A. Corn Predictive Outcomes

The Corn predictive architecture was evaluated using cyclic walk-forward validation on a dataset of 1,811 biologically complete county-year records across Texas. The model

achieved a cumulative out-of-time R^2 of 0.6774, with a Root Mean Square Error (RMSE) of 24.45 bu/ac and a Mean Absolute Error (MAE) of 19.50 bu/ac. These results indicate that the XGBoost framework, strengthened by engineered bio-climatic proxies, explained approximately 68% of the observed annual yield variability in Texas Corn.

Model performance varied across test years, reflecting differences in seasonal conditions and reporting consistency. The strongest yearly performance was observed in 2021 ($R^2 \approx 0.807$) and 2018 ($R^2 \approx 0.754$), with strong stability also maintained in 2020 ($R^2 \approx 0.7378$), 2022 ($R^2 \approx 0.658$), and 2024 ($R^2 \approx 0.659$). Lower performance occurred in 2019 ($R^2 \approx 0.305$), 2023 ($R^2 \approx 0.338$), and 2025 ($R^2 \approx 0.540$), suggesting that the model was less consistent in years with more abnormal weather patterns or noisier yield outcomes.

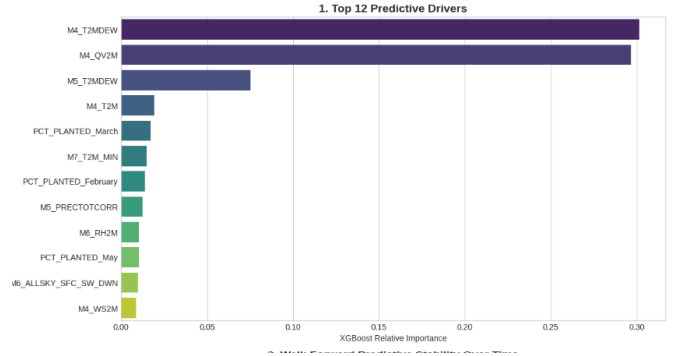


Fig. 1. Top 12 Predictive Drivers. A feature-importance plot showing the twelve most influential variables in the XGBoost model.

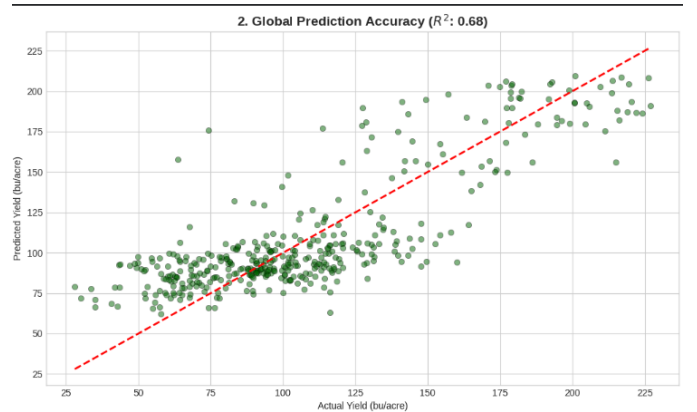


Fig. 2. Global Prediction Accuracy. A scatter plot comparing actual yield and predicted yield across the cumulative out-of-time test dataset, with a 1:1 parity line included for visual reference.

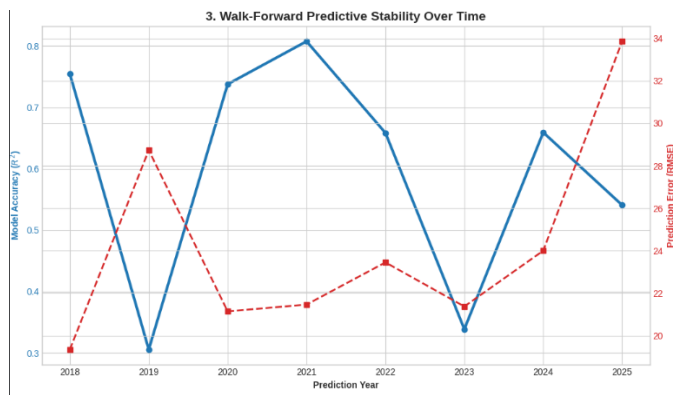


Fig. 3. Walk-Forward Predictive Stability Over Time. A dual-axis line plot showing annual model accuracy (R^2) and prediction error (RMSE) across walk-forward test years.

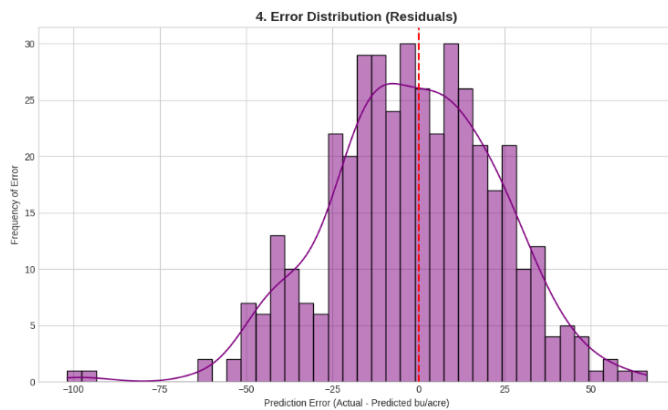


Fig. 4. Error Distribution (Residuals). A histogram of residual error, defined as actual yield minus predicted yield, showing the central tendency and spread of model error.

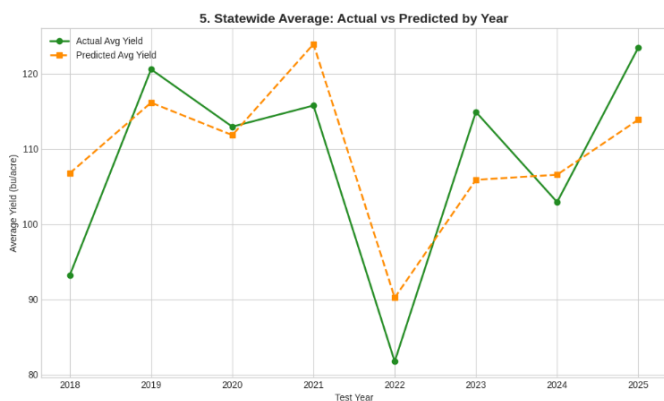


Fig. 5. Statewide Average: Actual vs Predicted by Year. A line plot comparing observed statewide mean yield and model-predicted statewide mean yield by year.

B. Cotton Predictive Outcomes

The Cotton model was evaluated using the same cyclic walk-forward validation framework on 1,380 biologically complete county-year records across Texas. In the current run, the available out-of-time test year was 2018, and the model achieved an R^2 of 0.5479, with an RMSE of 235.20 lbs/acre and an MAE of 200.13 lbs/acre. These values show that the model captured a meaningful share of Cotton yield variability, although with lower overall accuracy than the Corn system.

Because the present Cotton workflow produced only one out-of-time test year, the reported performance should be interpreted as a limited but still useful validation of the framework. Even so, the model output showed that the engineered variables were able to capture a real signal between weather, soil, and county-level Cotton yield. Overall, the Cotton results support the usefulness of the approach while also showing that Cotton was the more difficult crop to predict under the available data conditions.

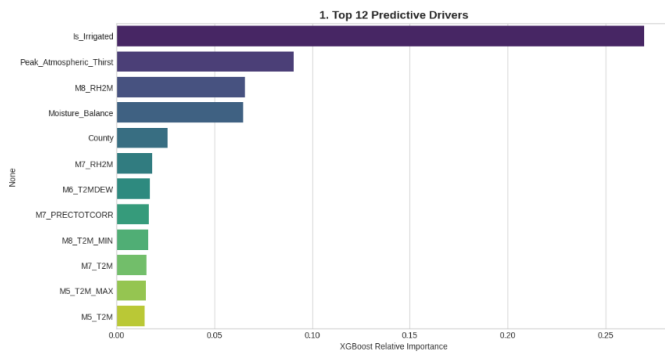


Fig. 6. Top 12 Predictive Drivers. A feature-importance plot showing the twelve most influential variables in the Cotton XGBoost model.

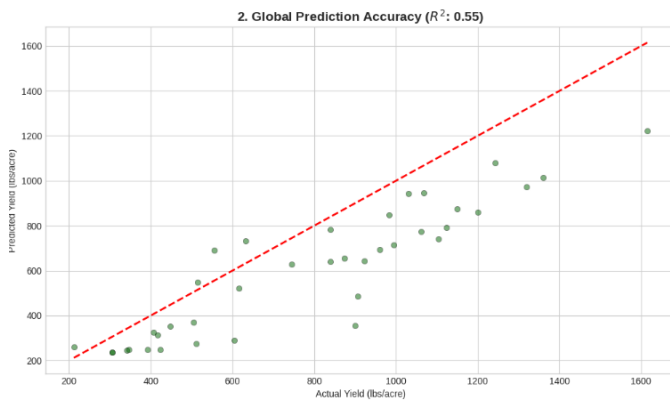


Fig. 7. Global Prediction Accuracy. A scatter plot comparing USDA-reported Cotton yield and XGBoost-predicted yield across the out-of-time test set, with a 1:1 parity line shown for reference.

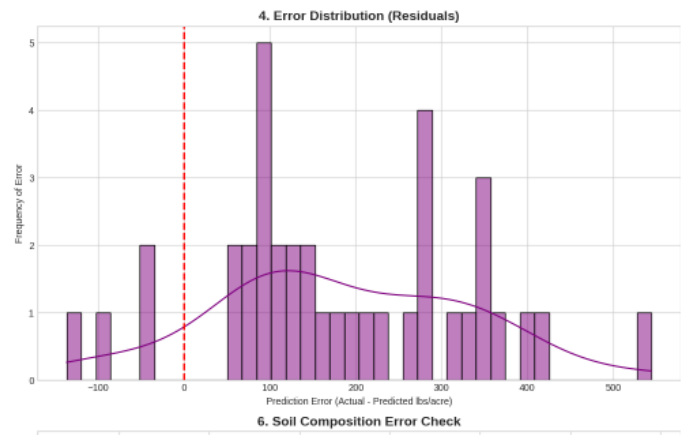


Fig. 8. Soil Composition Error Check. A scatter plot of residual error versus topsoil clay percentage, used to assess whether model error shifts systematically across soil texture conditions.

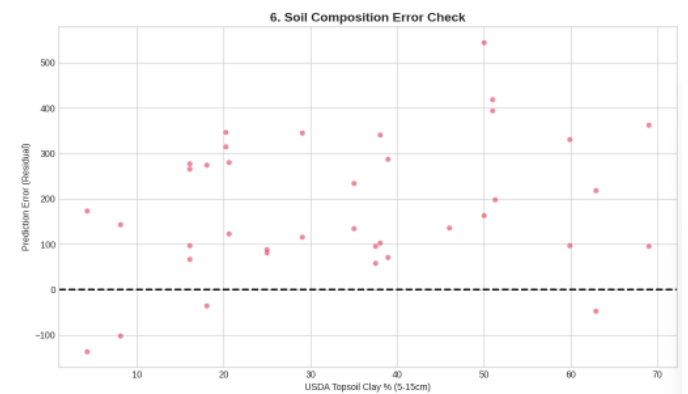


Fig. 9. Model Logic: Moisture Balance vs Predicted Yield. A scatter plot with trend line showing the relationship between the engineered Moisture_Balance variable and predicted Cotton yield.

IV. DISCUSSION

The predictive architecture demonstrated superior performance for Corn relative to Cotton. This divergence is primarily attributable to disparities in longitudinal data availability and the divergent physiological stress responses inherent to each crop. The Corn dataset benefited from a higher volume of biologically complete records and the successful implementation of a multi-year, cyclic walk-forward validation matrix. This enabled the algorithm to identify and validate highly stable predictive patterns across a diverse climatological timeline. Conversely, the Cotton analysis was constrained by systemic data sparsity; the merged county-level yield series was truncated post-2018, restricting the final workflow to a single out-of-time validation year. This statistical limitation inherently restricts the comprehensive evaluation of the Cotton model prior to accounting for biological complexities.

Furthermore, Corn presents a mathematically more direct predictive target due to its determinate growth cycle. Catastrophic yield degradation in Corn is strictly constrained to acute phenological windows; thus, extreme thermal or moisture deficits during these sensitive periods translate directly into final reported yields. This biological directness explains the stronger cumulative model accuracy, while also contextualizing its performance degradation during years of anomalous climatic extremes. For instance, the severe hydrological flooding disasters of 2019 and the catastrophic 2023 Heat Dome—which subjected dryland crops in regions such as the Rio Grande Valley to persistent temperatures exceeding 110°F—introduced localized volatilities. Such extreme, hyper-local events are inherently difficult to represent utilizing macro-level, county-wide monthly weather aggregations, logically explaining the model’s weakened precision during highly abnormal harvest years [3].

Cotton presents a fundamentally more complex modeling challenge due to its indeterminate growth habit. Final fiber yield is the cumulative result of a protracted chain of stress events, including prolonged thermal stress, atmospheric dryness, uneven precipitation, and late-season harvest conditions. Because Cotton possesses the biological capacity to partially recover from mid-season stress if subsequent conditions improve, the direct algorithmic mapping of specific meteorological anomalies to final yield is significantly attenuated. Consequently, while the model accurately predicts many individual yield values, it inherently loses consistency when weather patterns become highly irregular, resulting in a lower overall fit (R^2) compared to Corn.

Therefore, the statistical discrepancy between the two predictive targets must not be interpreted as a binary success or failure of the algorithmic framework. The Corn model, bolstered by deeper data reserves, multiple validation years, and a direct stress-to-yield physiological mechanism, serves as a robust validation of the foundational machine learning architecture. Conversely, the Cotton model was tasked with resolving a highly noisy biological problem under restricted data conditions. The successful capture of an authentic biolog-

ical signal despite Cotton’s indeterminate recovery mechanics and extreme weather volatility demonstrates the underlying resilience of the bio-climatic feature engineering methodology.

V. SUMMARY

This study built an interpretable machine learning framework for county-level yield prediction in Texas using USDA NASS yield data, NASA POWER weather data, soil variables, engineered bio-climatic proxies, and walk-forward validation. The goal was not just to predict yield, but to test whether XGBoost could capture non-linear weather-yield relationships while still remaining practical and explainable.

The results showed that the approach worked best for Corn, which produced the stronger and more stable validation, while Cotton still showed useful predictive value but under a more limited dataset and only one out-of-time test year in the final workflow. This suggests that model performance depended not only on algorithm choice, but also on crop physiology, dataset size, and how clearly seasonal stress appeared in the reported county-level yield.

More broadly, the findings support the idea that high-value agricultural prediction does not always require black-box deep learning models. XGBoost is a scalable tree-boosting approach that can deliver strong predictive performance with interpretability and practical speed, making it well suited for regional agricultural analytics. Overall, this study shows that traditional machine learning models are still valuable when they are built around crop biology, local environmental limits, and the realities of agricultural reporting.

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